Fractal–Spectral CIEL/0 Metatime Framework: From Arithmetic Foundations to Emergent Dynamics

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Abstract

We present a complete fractal–spectral formalism that generates particle masses, nuclear binding energies and low-energy observables from arithmetic and topological foundations. The framework integrates: (i) a fractal spectral kernel seeded by twin-primes and Collatz orbits, (ii) the first nontrivial zero of the Riemann zeta function as a spectral scale, and (iii) a single fundamental operator of information O_I whose canonical value and Heisenberg–like fluctuation are determined by the spectral structure. We give the Lagrangian, Hamiltonian and action principles; show how physical masses and binding energies emerge via minimization of the action; and list falsifiable predictions. A static-sector solver implementing the model (sampling O_I with the spectral fluctuation rule) is available at my GitHub.

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1 Introduction

Contemporary attempts to explain the pattern of fermion masses, nuclear binding, and cosmological parameters often rely on empirical input or large parameter spaces. Here we outline a theory in which these observables are emergent consequences of an arithmetic–geometric kernel: twin-primes, Collatz orbit structure, the golden ratio φ, and the first nontrivial zero of the Riemann zeta function $\mathrm{Im}(s_1) \approx 14.134725\ldots$. A single operator of information O_I and its spectral fluctuation provide the only input scale; all couplings and effective coefficients are derived from the fractal spectral structure.

2 Arithmetic and Spectral Foundations

2.1 Spectral scale and the information operator

Define the canonical information operator

$$O_I^{\mathrm{ref}} \;=\; 0.00917 \quad \text{(canonical value)} \tag{1}$$

and the deterministic spectral fluctuation scale (Heisenberg–type rule)

$$\sigma_{O_I} \;=\; \frac{O_I^{\mathrm{ref}}}{\mathrm{Im}(s_1)} \approx \frac{0.00917}{14.134725} \approx 6.49 \times 10^{-4}. \tag{2}$$

In the static sector solver we sample O_I from a narrow Gaussian $\mathcal{N}(O_I^{\mathrm{ref}}, \sigma_{O_I}^2)$ and compute ensemble means and standard deviations for all observables.

2.2 Twin–prime / Collatz kernel and fractal resonances

Twin-primes $(p, p{+}2)$ seed Collatz orbit lengths and define a discrete hierarchy of eigenvalues $\{\tau_f\}$ used to index fermion families. A fractal resonance function $R(n)$ (constructed from local Mandelbrot stability indices and Collatz orbit statistics) partitions the spectrum into bands:

$$B_1 : R \in [1.15, 1.25] \quad \text{(light fermions)}, \qquad B_2 : R \in [1.32, 1.42] \quad \text{(heavy fermions \& nuclei)}.$$

Topological indices λ_f (integers or rational ratios) come from the twin-prime/Collatz data and control leading power–law scalings.

3 Metatime geometry

3.1 Manifold and coordinates

The metatime manifold is a compact Kähler geometry combining:

$$\mathcal{M}_\tau \simeq \mathbb{C}P^n \times S^2 \times \mathcal{T}^m,$$

with local coordinates:

$$O_I \in \mathbb{R}_+, \qquad \tau \in \mathbb{R} \text{ (proper metatime)}, \qquad \gamma \in S^1 \text{ (Berry phase)}.$$

A Kähler potential $K(z, \bar{z}; O_I)$ defines the metric and curvature; spectral/topological data determine K up to canonical choices fixed by the arithmetic kernel.

4 Lagrangian and dynamical structure

We present a compact Lagrangian density capturing the essential sectors:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{grav}}[g] \;+\; \mathcal{L}_{O_I}[O_I] \;+\; \mathcal{L}_\psi[\psi, O_I] \;+\; \mathcal{L}_{\text{int}}[\psi, O_I, g]. \tag{3}$$

4.1 Gravitational sector

A spectrally modified Einstein–Hilbert action with O_I-dependent cosmological term:

$$\mathcal{L}_{\text{grav}} \;=\; \frac{1}{2\kappa^2}\sqrt{-g}\,\big(R - 2\Lambda(O_I)\big)\,, \qquad \Lambda(O_I) = \Lambda_0\,\mathcal{F}_\Lambda(O_I), \tag{4}$$

where $\mathcal{F}_\Lambda$ is a deterministic function derived from the spectral kernel (e.g. via Euler–Berry global constraint).

4.2 Information operator sector

A canonical kinetic term with fractal potential,

$$\mathcal{L}_{O_I} \;=\; \tfrac{1}{2}\sqrt{-g}\,Z(O_I)\,(\nabla O_I)^2 - \sqrt{-g}\,V_{\text{fractal}}(O_I), \tag{5}$$

where $V_{\text{fractal}}(O_I)$ is constructed as a spectral sum over twin-primes

$$V_{\text{fractal}}(O_I) \;=\; \sum_{p \in \text{twin-primes}} w_p\,\Phi_p(O_I)$$

with weights w_p and mode shapes Φ_p fixed by the arithmetic kernel (no free fitting parameters).

4.3 Fermion sector and emergent Yukawa couplings

Dirac fermions ψ_f acquire effective masses via topological couplings to O_I:

$$\mathcal{L}_\psi \;=\; \sum_f \bar{\psi}_f\big(i\not{D} - m_f(O_I)\big)\psi_f, \qquad m_f(O_I) = M_0\,\lambda_f^\beta\, e^{C_f(O_I - O_I^{\rm ref})}. \tag{6}$$

Here λ_f are topological/hierarchical integers from twin-prime data, $\beta = 1/(1+\mathrm{Im}(s_1)/10) \approx 0.414$ encodes spectral saturation, and C_f are fractal indices computed from local fractal dimension and linking numbers (in the completed theory C_f are derived, not fitted).

Interaction terms are *emergent Yukawa couplings* (not elementary Higgs):

$$\mathcal{L}_{\rm int} \supset -\sum_f y_f(O_I)\,\bar{\psi}_f\psi_f\,O_I\,, \qquad y_f(O_I) \equiv \tilde{y}_f\, e^{C_f(O_I - O_I^{\rm ref})}.$$

All functions above are fixed by the arithmetic–fractal kernel and $\mathrm{Im}(s_1)$.

5 Hamiltonian and field equations

Legendre transform of (3) yields canonical momenta and a Hamiltonian density $\mathcal{H}$. Euler–Lagrange equations produce:

$$\frac{\delta S}{\delta O_I} = 0 \quad\Rightarrow\quad \Box\big(Z(O_I)O_I\big) + V'_{\rm fractal}(O_I) + \sum_f \frac{\partial m_f(O_I)}{\partial O_I}\bar{\psi}_f\psi_f + \cdots = 0, \tag{7}$$

$$\frac{\delta S}{\delta \bar{\psi}_f} = 0 \quad\Rightarrow\quad (i\not{D} - m_f(O_I))\psi_f = 0, \tag{8}$$

$$\frac{\delta S}{\delta g^{\mu\nu}} = 0 \quad\Rightarrow\quad G_{\mu\nu} + \Lambda(O_I)g_{\mu\nu} = \kappa^2 T_{\mu\nu}. \tag{9}$$

6 Emergent hierarchy and vacuum

The vacuum expectation value $\langle O_I\rangle$ is determined by spectral minimization:

$$\left.\frac{\partial V_{\rm fractal}}{\partial O_I}\right|_{O_I = \langle O_I\rangle} = 0,$$

which (with the arithmetic kernel specified) yields $\langle O_I\rangle \approx O_I^{\rm ref}$. Physical fermion masses then follow

$$m_f^{(0)} = M_0\,\lambda_f^\beta\,,$$

and small fluctuations around $\langle O_I\rangle$ produce tiny, deterministic spread set by σ_{O_I}.

7 Static-sector solver and sampling rule

The static-sector solver implements the deterministic sampling rule

$$O_I \sim \mathcal{N}\big(O_I^{\mathrm{ref}}, \sigma_{O_I}^2\big), \qquad \sigma_{O_I} = \frac{O_I^{\mathrm{ref}}}{\mathrm{Im}(s_1)}.$$

Ensemble means and standard deviations for masses and binding energies are reported as the model predictions. The solver implementing the static sector is available at: LINK.

8 Predictions and falsifiable tests

The framework yields the following sharpened predictions (all numbers are ensemble means; uncertainties derive from σ_{O_I} sampling):

- **Neutrino sector:** Normal ordering with masses

  $$m_{\nu_1} \approx 4.22 \times 10^{-4}\ \mathrm{eV}, \quad m_{\nu_2} \approx 6.41 \times 10^{-3}\ \mathrm{eV}, \quad m_{\nu_3} \approx 5.02 \times 10^{-2}\ \mathrm{eV},$$

  $\sum_i m_{\nu_i} \approx 0.0571$ eV (within current cosmological bounds).

- **DUNE resonance structure:** CP-resonant enhancement at $E_0 = 0.63$ GeV with width ± 50 MeV and secondary modes at 0.21 and 0.09 GeV. (Falsification: absence of a resonance at $E_0 \pm 50\,\mathrm{MeV}$ at $\geq 3\sigma$.)

- **CMB low-ℓ power:** enhancement factor ≈ 2.7 for $\ell \lesssim 100$ relative to ΛCDM expectation (Simons-class sensitivity).

- **Accordion BEC:** multi-site soliton phase sum magnitude $|\sum_j e^{i\phi_j}| < 0.11$.

- **Large-scale structure (Euclid):** modulation of galaxy correlation at scale $r \approx 100$ Mpc with amplitude $\sim 7.5\%$.

Each prediction is falsifiable with current or near-term experiments; full statistical criteria are implemented in the solver's validation suite.

9 Predicted masses and binding energies (tables)

The tables below list ensemble mean predictions (static-sector solver) together with reference experimental values used for validation.

10 Discussion and outlook

We have shown how a compact arithmetic–fractal kernel combined with a single information operator O_I and a spectral fluctuation rule generates a tight, falsifiable framework for particle and nuclear physics. Remaining tasks to push this program from a complete formalism to a widely accepted theoretical foundation include:

Table 1: Charged lepton masses — model prediction (ensemble mean) vs. PDG reference.

Particle	Model prediction (MeV)	PDG (MeV)
e	0.5109989461	0.5109989461
μ	105.6583745	105.6583745
τ	1776.86	1776.86

Table 2: Quark effective masses used in the static sector (MeV). These are the ensemble-mean outputs of the static solver (reference masses used as the baseline).

Quark	Model prediction (MeV)	PDG nominal value (MeV)
u	2.16	2.16
d	4.67	4.67
s	93.5	93.5
c	1270	1270
b	4180	4180
t	173000	173000

1. A fully explicit derivation of the fractal indices C_f and topological integers λ_f from first-principles spectral geometry (twin-prime $\rightarrow$ linking numbers $\rightarrow$ fractal dimensions).
2. Non-perturbative solution of the O_I field equations (time evolution, scattering amplitudes, decay rates).
3. More detailed matching to QCD in the hadronization regime (derive effective hadronic Lagrangian from the fermion + O_I sector).

The static-sector solver (sampling rule and ensemble predictions) is provided as a reproducibility artifact.

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References

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[2] M. V. Berry, "Quantal phase factors accompanying adiabatic changes", Proc. R. Soc. Lond. A **392**, 45–57 (1984).

[3] A. Lipa, "Metatime Framework", Github (2025/26) github.com/AdrianLipa90/metatime

Table 3: Neutrino masses — model predictions (eV).

State	Model prediction (eV)	Comment
m_{ν_1}	4.22×10^{-4}	lightest state
m_{ν_2}	6.41×10^{-3}	solar splitting consistent
m_{ν_3}	5.02×10^{-2}	atmospheric splitting calibrated
$\sum_i m_{\nu_i}$	5.71×10^{-2}	cosmological bound $\lesssim 0.12$ eV

Table 4: Nucleon masses and nuclear binding per nucleon — model vs. experimental references.

Observable	Model mean	Model 1σ	Reference	Source
m_p (MeV)	938.27	$\mathcal{O}(10^{-1}$ MeV)	938.27208816	PDG 2024
m_n (MeV)	939.57	$\mathcal{O}(10^{-1}$ MeV)	939.56542052	PDG 2024
Binding per nucleon (O-16)	7.976 MeV	< 0.01 MeV	7.976 MeV	empirical
Binding per nucleon (Fe-56)	8.790 MeV	< 0.05 MeV	8.790 MeV	empirical
Binding per nucleon (Pb-208)	7.867 MeV	< 0.05 MeV	7.867 MeV	empirical